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Jet measurements in Oxygen-Oxygen (O+O) collisions at $\sqrt{s_{\mathrm{NN}}} = 200$ GeV with the sPHENIX detector

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Abstract

This analysis note presents the measurement of inclusive jet production in oxygen-oxygen (O+O) collisions at a nucleon-nucleon center-of-mass energy of $\sqrt{s_{\mathrm{NN}}}$ = 200 GeV, collected with the sPHENIX detector in 2026. The dataset corresponds to an integrated luminosity of 52.4 nb^{-1}, recorded using a jet trigger in coincidence with the minimum-bias detector (MBD). Jets are reconstructed from the electromagnetic (EMCal) and hadronic (HCal) calorimeters using the anti-k_t algorithm with a jet radius of $R = 0.4$. The jet central-to-peripheral ratio (R_{CP}) and nuclear modification factor (R_{AA}) are reported as a function of transverse momentum (p_{T}) within pseudorapidity $|\eta| < 0.7$.

Contents

1 Introduction

One of the central goals of high-energy heavy ion collisions is to characterize the Quark-Gluon Plasma (QGP), the strongly coupled QCD matter formed at extreme temperatures and energy densities. O+O collisions occupy an unique position in the system size landscape, bridging the QGP regime of large nucleus-nucleus (A+A) collisions (Au+Au) and small systems (p+p and p+A). A powerful probe to study the properties of the produced medium is jet quenching, the suppression of jet yields in A+A collisions.

This suppression can be experimentally quantified by comparing the produced jet yields in A+A to that in p+p (R_{AA}) or to peripheral A+A collisions (R_{CP}), both scaled by the number of binary nucleon-nucleon collisions (N_{coll}). The R_{AA} is defined as

$$R_{AA} = \frac{1}{A^2}\frac{d^2\sigma^{AA}/dp_T d\eta}{d^2\sigma^{pp}/dp_T d\eta}, \tag{1}$$

where A is the mass number of the oxygen nucleus. The central-to-peripheral ratio, R_{CP}, is defined as

$$R_{CP} = \frac{\frac{1}{N_{coll}}\frac{d^2N_{jet}}{dp_T d\eta}|_{central}}{\frac{1}{N_{coll}}\frac{d^2N_{jet}}{dp_T d\eta}|_{peripheral}}, \tag{2}$$

with N_{jet} denoting the number of jet yields in a given kinematic interval.

Predictions are available in the literature for R_{AA} in minimum bias O+O collisions at 200 GeV without quenching effects [1], providing a key baseline for these measurements. At RHIC, the STAR experiment has submitted for publication results on jet quenching in O+O collisions [2]. STAR shows measurements of correlations between charged hadron triggers of high transverse momenta ($7 < p_{\mathrm{T}} < 30$ GeV) with recoiling charged hadrons ($3 < p_{\mathrm{T}} < 7$ GeV) or charged–particle jets ($p_{\mathrm{T}_{jet}} > 8$ GeV) in event–activity selected O+O collisions. At the LHC, the CMS experiment has published charged hadron R_{AA} in O+O at 5.36 TeV [3] indicating significant jet quenching beyond cold nuclear matter effects. The ATLAS experiment has preliminary results on the dijet asymmetry in O+O compared with p+p collisions, where a modestly increase dijet imbalance is observed in central O+O collisions [4]. No inclusive jet results have been published at RHIC or the LHC to date.

The dataset for this analysis comprises O+O collisions at $\sqrt{s_{\mathrm{NN}}}$ = 200 GeV taken in 2026, with a jet trigger (Jet12) in coincidence with the minimum-bias detector (MBD). The online energy threshold for the trigger was set to 12 GeV. Jets are reconstructed from electromagnetic (EMCal) and hadronic (HCal) calorimeters using the anti-k_t algorithm for R = 0.4. The kinematic range of the jets is $19 < p_{\mathrm{T}} < 70$ GeV and $|\eta| < 0.7$. The underlying event (UE) is estimated event-by-event and is subtracted for each measured jet.

The note is structured as follows. Section 2 describes the data and Monte Carlo (MC) samples, the trigger, event selections, and reweighting procedures on the MC. Section 4 describes the jet selection and background removal procedure. Section 5 details the jet energy scale (JES) and jet energy resolution (JER) in this analysis compared to those in the inclusive jet measurement analysis in p+p [5]. The UE is described in Section 6. Section 7 details the overall unfolding procedure including the purity and efficiency corrections. Section 8 describes the systematic uncertainties. Finally, Section 9 presents the results.

2 Dataset and Event Selection

This section details the data and Monte Carlo samples utilized in this analysis, as well as the trigger and event selection.

2.1 Data and MC

2.1.1 Data

The data sample used for this analysis was collected in 2026 with O+O collisions at $\sqrt{s_{\mathrm{NN}}}$ = 200 GeV. The running conditions was almost entirely at a crossing angle of 1.5 mrad, and thus the z-vertex distribution was reasonably narrow. Prior to selections, the real data monitoring of integrated luminosity is shown in Figure 1.

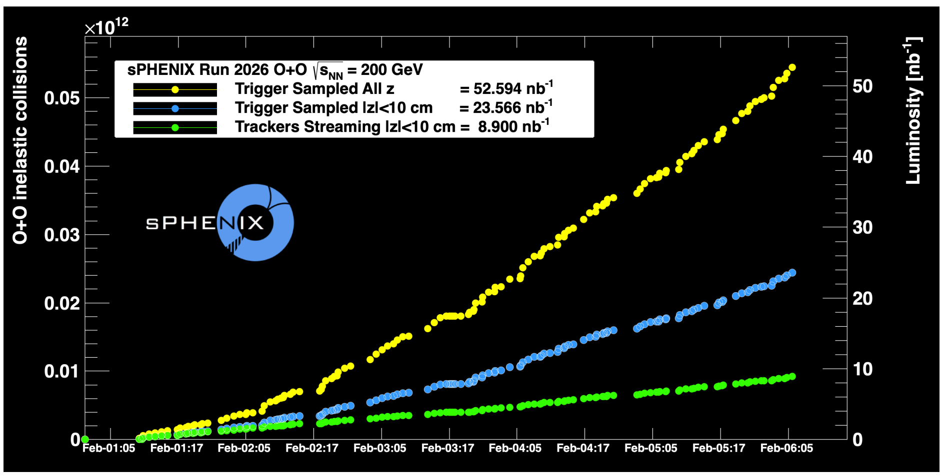

Figure 1: Real-time sPHENIX luminosity counting during the 2026 O+O data taking period.

An offline quality assurance (QA) procedure has been applied to select a set of "golden" runs for calorimeter-based analyses, following the same criteria established for the sPHENIX $p+p$ and Au+Au programs:

- Pass HCal QA
- Pass EMCal QA
- Pass MBD vertex QA
- Pass trigger & Data Acquisition QA

The details of the creation of the run list can be found in this link. The reconstruction production and calibration tags are `pro001_pcdb001_v001` and `new_newcdbtag`, respectively.

2.1.2 MC

This analysis uses PYTHIA-8 [6] generated jets embedded into O+O HIJING [7] events. A minimum bias HIJING O+O sample is also used for event characterization and reweighting to data. To optimize statistics across the studied p_{T} range, the PYTHIA-8 jet samples are generated with different $\hat{p}_T^{\mathrm{min}}$ values and leading truth jet p_{T} selections, as summarized in Table 1.

Sample	Configuration	$\hat{p}_T^{\mathrm{min}}$ [GeV]	Truth jet p_{T} range [GeV]	Cross section [barn]	Events (M)
MB	minbias	–	–	4.197×10^{-2}	5
Jet5	minbias	–	7–14	1.3878×10^{-4}	5
Jet12	minbias	–	14–21	1.4903×10^{-6}	5
Jet20	HardQCD+PromptPhoton	10	21–32	6.2623×10^{-8}	5
Jet30	HardQCD+PromptPhoton	15	32–42	2.5298×10^{-9}	5
Jet40	HardQCD+PromptPhoton	22	42–52	1.3553×10^{-10}	5
Jet50	HardQCD+PromptPhoton	29	>52	7.3113×10^{-12}	5

Table 1: Configuration, cross sections, and number of events for each of the PYTHIA-88 MC inclusive jet samples used in this analysis.

2.2 Event Selection

Events are selected using the **Jet12** trigger, which requires an online calibrated energy of at least 12 GeV on the L1 jet trigger primitive, in coincidence with at least one hit on each side of the MBD (north and south).

For the offline analysis, the following selection criteria are applied:

- Reconstructed MBD z-vertex within $|v_z| < 60$ cm
- At least one MBD PMT with charge > 0.5 in each north and south

This offline selection removes 7.5% of the total events and is applied as a correction in the luminosity calculation, yielding a total integrated luminosity of 48.5 nb^{-1}.

2.3 Pileup Effects

The impact of pileup is investigated using a template fit to the MBD charge sum distribution of minimum-bias (MB) triggered events. The MB trigger requires at least one hit on each MBD north and south sides, with an online z-vertex cut to be within [-150,150] cm. The single-collision template is constructed from a dedicated low-luminosity fill. Templates for double and triple collisions are then generated by randomly sampling events from the single-collision template. Figure 2 shows the MBD charge sum distribution of MB triggered events from nominal-luminosity runs, fitted with a sum of the three templates. The pileup contribution is found to be less than 1% of the total events and hence neglected for this analysis.

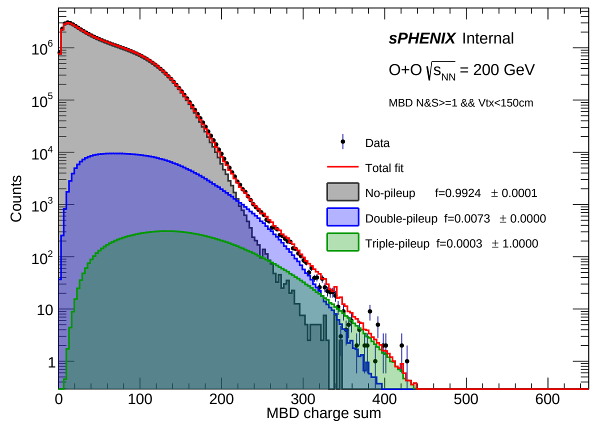

Figure 2: The MBD charge sum distribution of MB triggered events, fitted with the sum of single, double, and triple interaction templates.

2.4 Centrality Calibration

The centrality in O+O collisions is determined by fitting the MBD charge sum to a convolution of MC Glauber and negative binomial distribution (MC Glauber $\oplus$ NBD) using a minimum-bias trigger. Figure 3 left and right panels show the fits of the MC Glauber $\oplus$ NBD to data and MC, respectively.

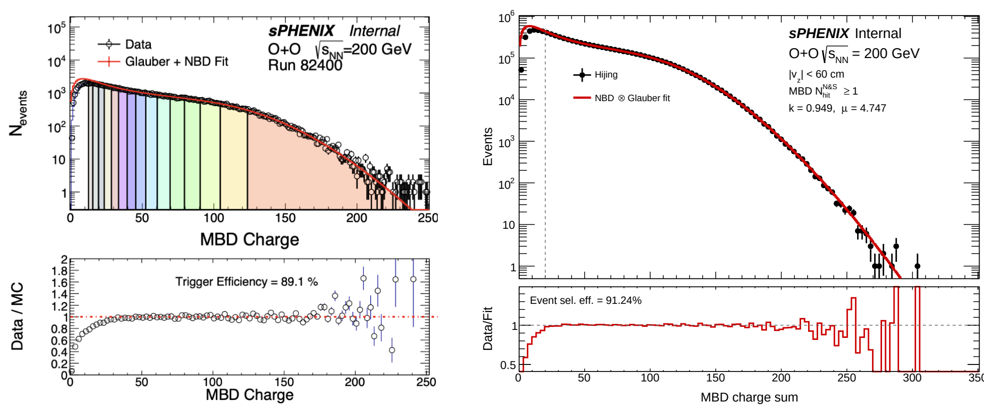

Figure 3: The MBD charge sum distribution in minimum-bias O+O events (points), compared to the best fit from a MC Glauber $\oplus$ NBD simulation (red line) for data (left) and Hijing MC (right).

The MC Glauber for O+O collisions has been extensively studied for LHC energy collisions [8]. Extending those studies including Oxygen nucleus configuration variations to RHIC energies is shown in Figure 4. The total inelastic O+O cross section is quoted as $\sigma_{OO} = 1.15 \pm 0.05$ barns.

During the O+O data taking period, a Vernier Scan was carried out under the expert guidance of Angelika Drees (C-AD). The ZDCNS, i.e., at least one neutron in the ZDC north and south, yielded a cross section of 0.57 barns (no uncertainty available yet). It is notable that in Au+Au collisions, the ZDCNS cross section is larger than the MBDNS cross section, due to the additional UPC and dissociative processes contributing to the ZDCNS. In the O+O case, these additional

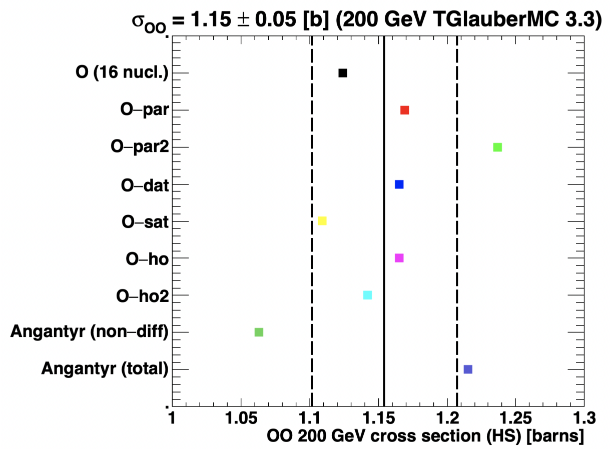

Figure 4: MC Glauber calculation results for the O+O cross section. the variations include different configurations of the nucleons in the Oxygen nucleus – as detailed in Ref. [8].

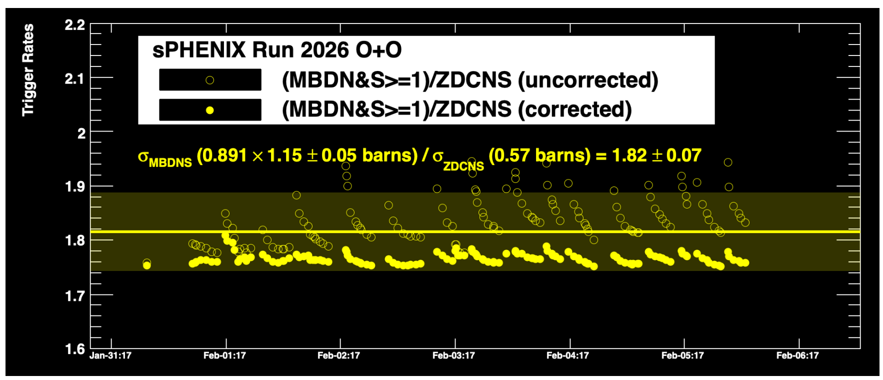

Figure 5: Ratio of MBDNS to ZDCNS trigger rates as a function of time during the run. The open points are uncorrected and the closed points are corrected for pileup contributions. The horizontal line and uncertainty box represent the expectation for the condition of MBDNS firing on 89.1% of the O+O inelastic cross section.

processes are a very small contribution and many inelastic O+O collisions do not have sufficient spectators to yields neutrons on both ZDC sides.

Figure 5 shows the online ratio of triggers for MBDNS and ZDCNS as a function of time. The open points are uncorrected and show a clear store dependence. The contributions from pileup are then corrected and the results shown as filled points. If the MBDNS fires of 89.1% of the O+O inelastic cross section of 1.15 barns, the expected ratio should be 1.82 ± 0.07 which is shown as a horizontal line and shaded band. The solid points agree within the stated uncertainties.

Figure 6 shows the run-by-run QA of the fraction of events in the 0–20% centrality interval passing the **Jet12** trigger, demonstrating the stability of the MBD and centrality calibration throughout the running period.

Plan to use the official centrality calibration once the calibration is in place.

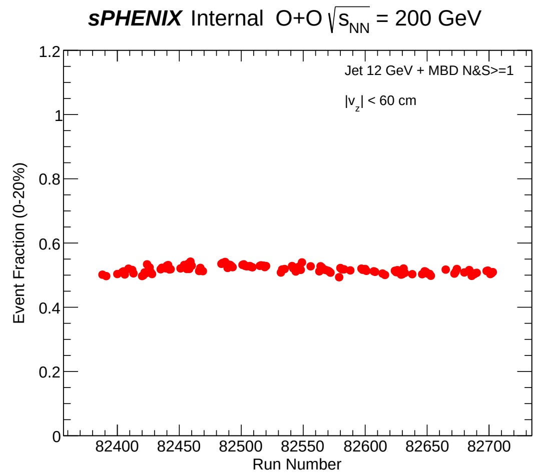

Figure 6: The fraction of events for 0–20% centrality interval as a function of run number in the jet12 triggered events.

3 MC Reweighting

To ensure realistic event characterization in the simulation, the MBD z-vertex and total calorimeter transverse energy (E_T) distributions in MC are reweighted to match those in data. Figure 7 compares the MBD v_z and total calorimeter E_T distributions between data and MC. Because the E_T in MC has an offset from data, it is rescaled first and then reweighted so that data and MC agree across the full E_T range. The resulting weight factors of v_z and E_T are applied event-by-event in MC.

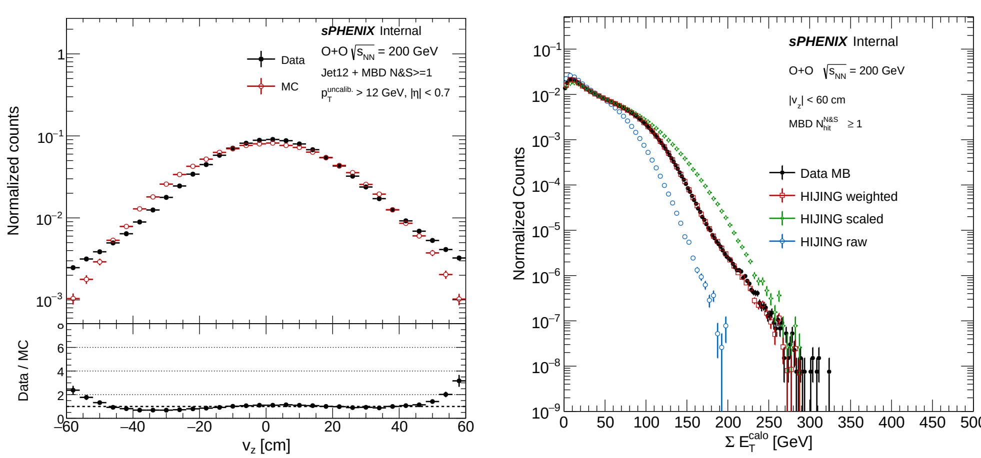

Figure 7: Comparisons of the MBD v_z (left) and total E_T (right) for data and MC

4 Jet Reconstruction and Background Rejection

Jets are reconstructed from EMCal and HCal towers using the anti-k_t algorithm with $R = 0.4$. To reject beam-induced and out-of-time cosmic backgrounds, the jet time is used as a discriminating variable. The jet time is defined as the energy-weighted mean time of the constituent towers. This variable was studied in the JetTG-TF01 team and more detailed studies can be found in the internal note [9]. To take into account the global time shift from the satellite bunch crossings, the difference between the jet time and the MBD mean time ($t^{\mathrm{jet-MBD}}$) is used for this analysis.

To characterize the timing distribution of background-free clean jets, a back-to-back dijet selection is used as an event-level tag. Events are required to contain at least one dijet pair that satisfies the following criteria:

- $\Delta\phi_{12} > 3\pi/4$
- $E_2/E_1 > 0.2$,

where the subscripts (1, 2) denote the leading and subleading jets in the event, respectively. Figure 8 shows the $t^{\mathrm{jet-MBD}}$ distribution before (left) and after (right) the selection of dijet-taggged events.

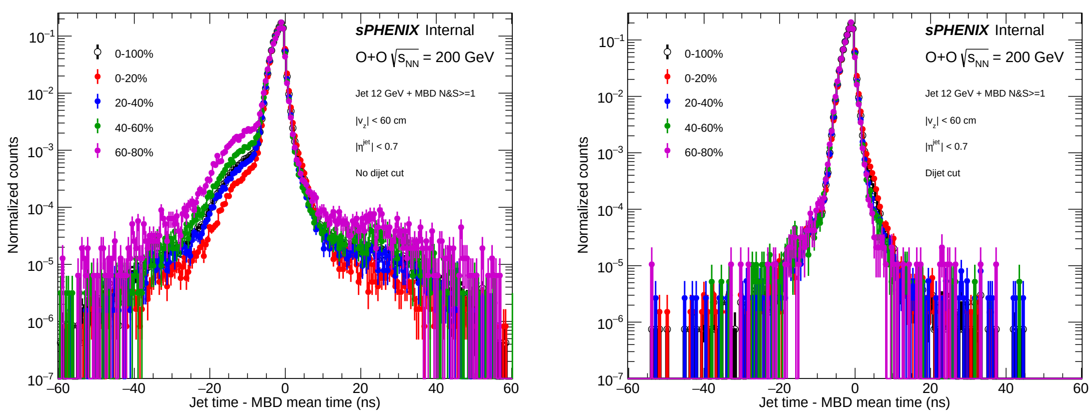

Figure 8: $t^{\mathrm{jet-MBD}}$ without (left) and with (right) the dijet-tagged event selection. The different colored points indicate each centrality intervals (0–20%, 20–40%, 40–60%, 60–80%, and 0–100%).

Because the timing response depends on the jet energy fraction in the EMCal and HCal, the dependence of $t^{\mathrm{jet-MBD}}$ is studied versus the jet EMCal energy fraction. Figure 9 shows the 2D correlation of $t^{\mathrm{jet-MBD}}$ and jet EMCal energy fraction for $25 < p_{\mathrm{T}} < 30$ GeV, with and without the dijet selection. Following the procedure of JetTG-TF01 [9], a jet EMCal energy fraction dependent correction is derived from a fit to the profile of this correlation and applied to each measured jet.

The left panel of Figure 10 shows the profiles of these correlations for dijet-tagged events in various jet p_{T} intervals. Because the profiles are found to be consistent across the p_{T} intervals, the correction is derived from the p_{T}-integrated profile ($p_{\mathrm{T}} > 19$ GeV) and commonly applied to all measured jets p_{T}. The right panel shows the fit of this profile using a third-order polynomial fit.

The correction function is applied to all measured jets. Figure 11 shows the corrected time,

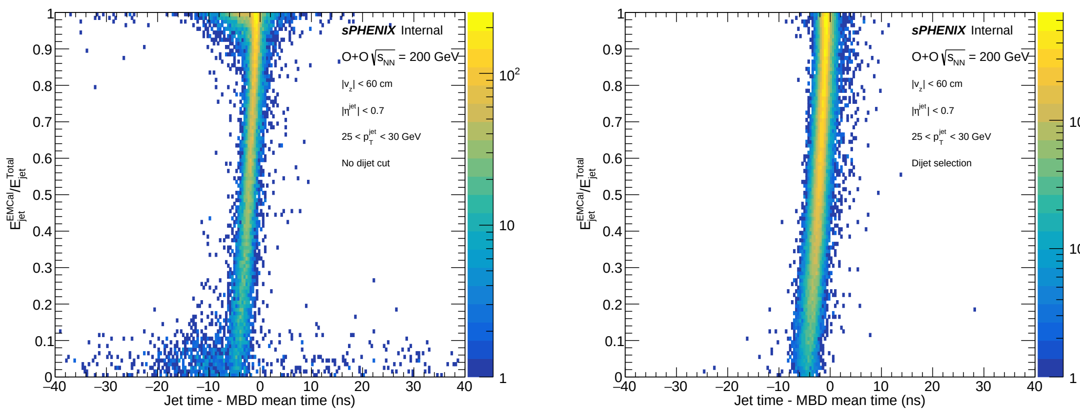

Figure 9: Jet EMCal energy fraction versus $t^{\text{jet}-\text{MBD}}$ without (left) and with (right) the back-to-back dijet selection, for $25 < p_{\text{T}} < 30$ GeV.

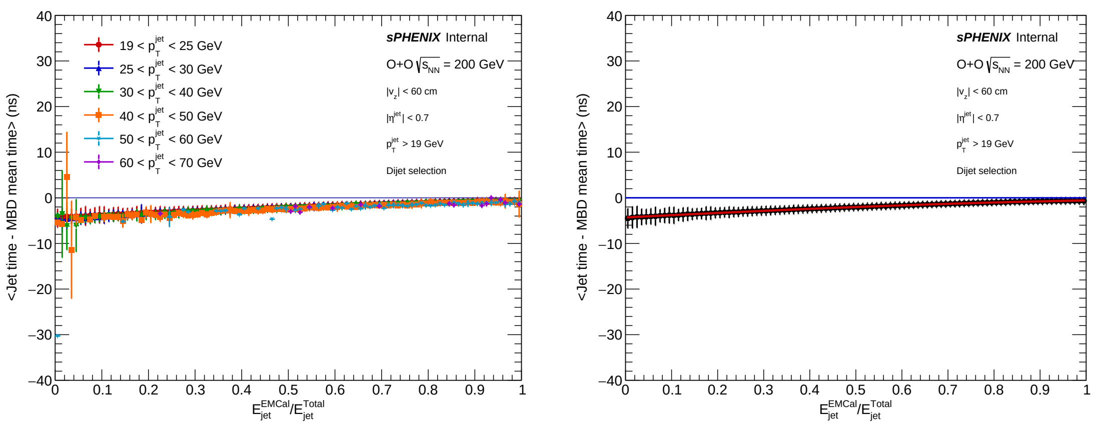

Figure 10: Profile of the EMCal energy fraction as a function of $t^{\text{jet}-\text{MBD}}$ for dijet-tagged events overlaid for different jet p_{T} intervals (left) and for $p_{\text{T}} > 19$ GeV (right). The red curve in the right panel is the third-order polynomial fit.

$t_{\text{corr}}^{\text{jet}-\text{MBD}}$, as a function of the jet EMCal energy fraction with and without the dijet selection for $25 < p_{\text{T}} < 30$ GeV.

The nominal jet selection requires $|t_{\text{corr}}^{\text{jet}-\text{MBD}}| < 2$ ns. Figure 12 shows the efficiency of this selection as a function of jet p_{T}, evaluated for dijet-tagged events. These efficiencies are applied as bin-by-bin corrections to the measured jet yields.

Figure 13 shows the jet p_{T} spectra after applying the $|t_{\text{corr}}^{\text{jet}-\text{MBD}}| < 2$ ns cut for different centrality classes.

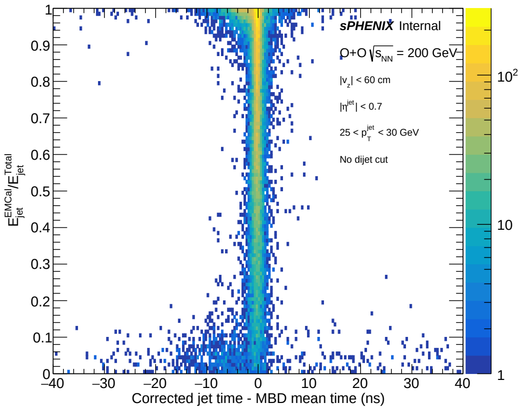

Figure 11: Corrected $t_{\mathrm{corr}}^{\mathrm{jet-MBD}}$ as a function of jet EMCal energy fraction without (left) and with (right) the dijet-tagged event selection, for $25 < p_{\mathrm{T}} < 30$ GeV.

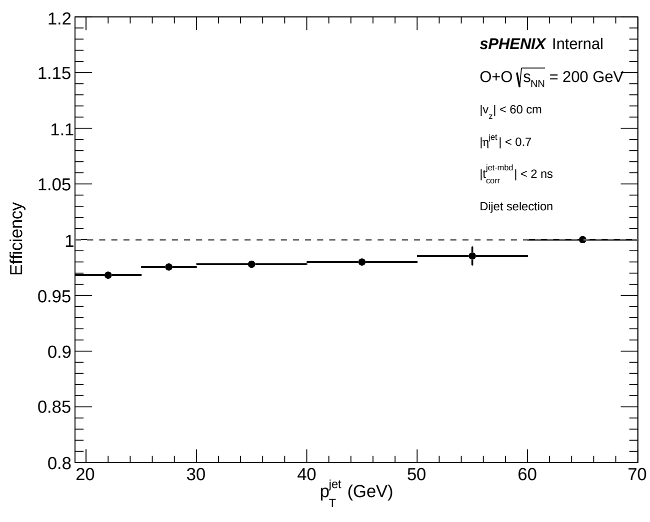

Figure 12: Efficiency of the $|t_{\mathrm{corr}}^{\mathrm{jet-MBD}}| < 2$ ns selection as a function of jet $p_{\mathrm{T}}$for dijet-tagged events.

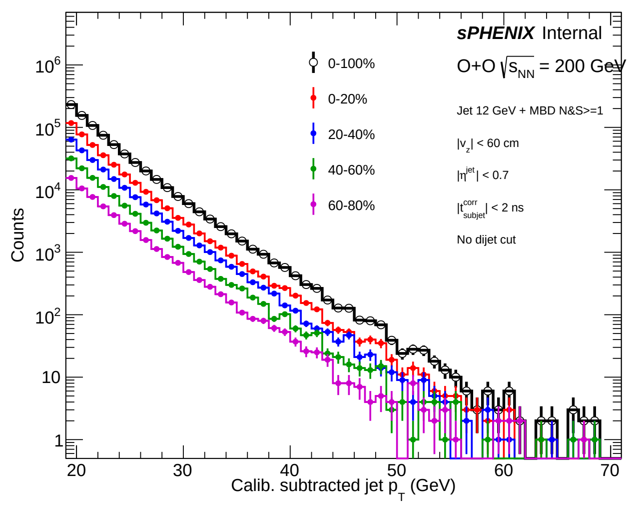

Figure 13: Jet p_{T} distribution for centrality intervals (0–20%, 20–40%, 40–60%, 60–80%, 0–100%).

5 Jet Energy Scale (JES) and Jet Energy Resolution (JER)

The jet energy scale (JES) and jet energy resolution (JER) are derived using data-driven methods in $p+p$ collisions from Run 2024 data [10, 11]. To assess their applicability in O+O, the JES and JER are studied in O+O MC and compared to those in $p+p$ MC, with the $p+p$-derived JES correction applied to jets in O+O.

For each p_T^{truth} interval, the p_T^{reco}/p_T^{truth} distribution is fitted with a Gaussian function, and the mean and width are extracted. This procedure is performed in fine p_T^{truth} intervals for each centrality interval and compared to those obtained in $p+p$. Figure 14 shows an example fit for a p_T^{truth} interval.

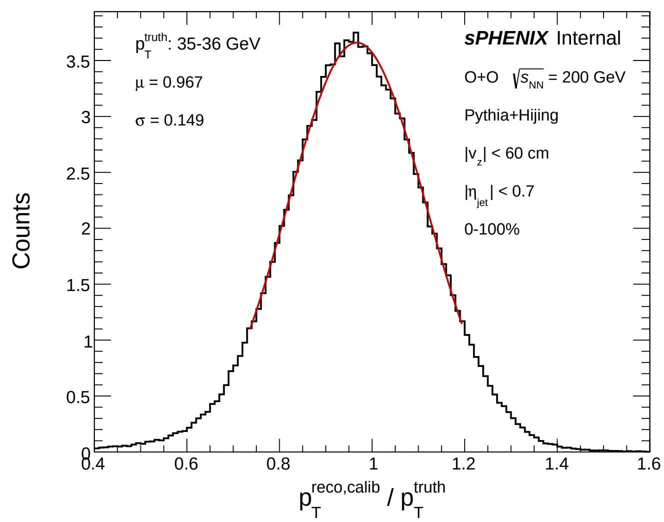

Figure 14: p_T^{reco}/p_T^{truth} distribution fitted with a Gaussian function for $35 < p_T^{truth} < 36$ GeV and centrality 0–100%.

Figure 15 shows the extracted mean (left) and width (right) of the p_T^{reco}/p_T^{truth} distribution as a function of p_T^{truth}, overlaid with the corresponding $p+p$ results with and without UE subtraction applied.

The O+O results are in good agreement with $p+p$ that include UE subtraction, demonstrating that the *in-situ* JES and JER calibrations derived in $p+p$ are applicable to O+O.

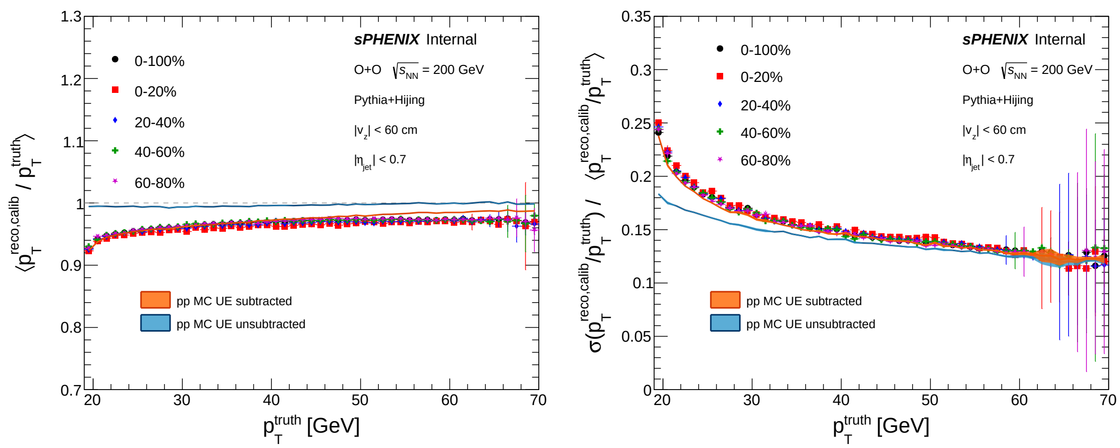

Figure 15: Mean (left) and width (right) of the p_T^{reco}/p_T^{truth} distribution as a function of p_T^{truth}. The orange and blue curves represent the results for p+p with and without UE subtraction, respectively.

6 Underlying Event (UE) Estimation

The underlying event (UE) subtraction uses an iterative method [12] in which the UE is estimated separately for each calorimeter layer in $\Delta\eta = 0.1$ strips.

First, the initial seeds are identified as $R = 0.2$ jets in which the leading constituent tower energy is greater than 3 GeV and its transverse energy is greater than three times the average constituent transverse energy. Then, For each η strip, the initial UE per tower is computed as the average tower energy in that strip, excluding towers within $\Delta R < 0.4$ of any initial seed jet.

In the second iteration, this initial UE is subtracted from each calorimeter tower and a new set of seed of $R = 0.2$ jets ($p_{\mathrm{T}} > 7$ GeV) is reconstructed from the subtracted towers. The UE per η strip for each calorimeter layer is recalculated with the second-iteration seeds excluded. This UE is defined as the final UE.

The towers, with the final UE subtracted, are then used as input to the anti-k_t algorithm to reconstruct the UE-subtracted jets used in this analysis.

Figure 16 shows the total UE summed over the three calorimeter layers as a function of the total calorimeter transverse energy (left) and the MBD charge sum (right), for both data and MC. The UE is consistent between data and MC over the full studied range of both variables, verifying the validation of the UE estimation in MC simulation.

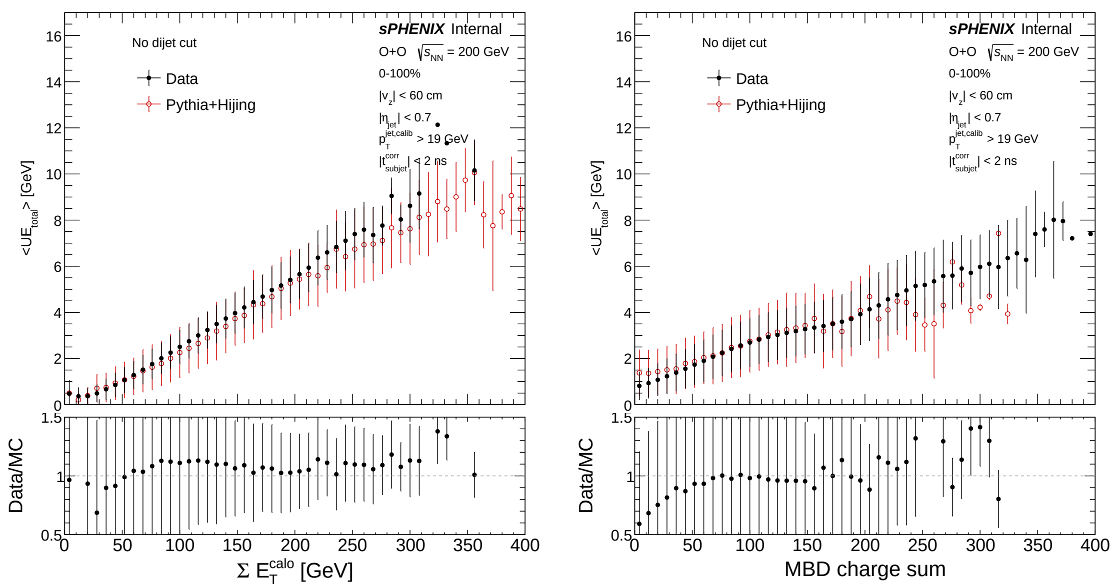

Figure 16: Total UE of the three calorimeter layers as a function of the total calorimeter transverse energy (left) and the MBD charge sum (right). The black and red points correspond to data and MC, respectively.

Figure 17 shows the total UE as a function of jet p_{T}. As expected from the higher event activity in central collisions, the results show a clear ordering from central to peripheral. Also, the UE is found to be independent of the jet p_{T}, as expected because the jets are excluded from the UE estimation.

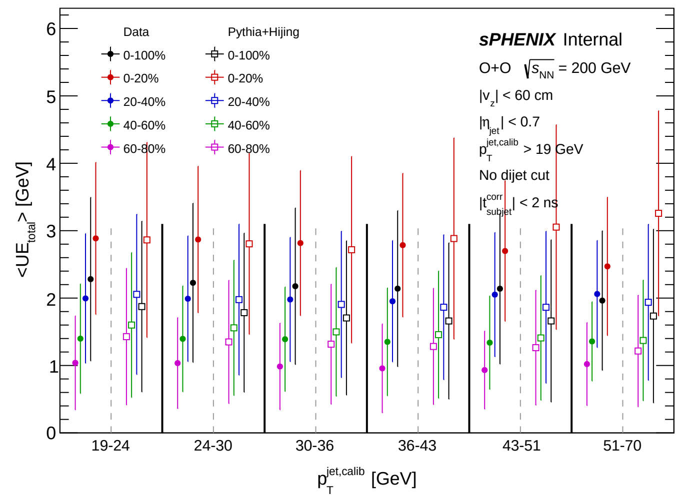

Figure 17: Total UE of all three calorimeter layers as a function of jet p_{T}. The filled and open markers correspond to data and MC, respectively.

7 Unfolding

The unfolding is performed with the RooUnfold package using the iterative Bayesian (D'Agostini) method. The inputs are the purity-corrected data spectrum, the prior distribution, and the response matrix. Truth and reco jets are matched within $\Delta R < 0.3$. Unmatched reco and truth jets are categorized as "fake" and "miss," and are corrected for using purity and efficiency, respectively.

In order to match the MC p_{T} spectrum to that in data, the prior is reweighted using a single-iteration unfolding of the data with the unweighted baseline MC simulation. Figure 18 compares the resulting unfolded p_{T} spectrum from data to the unweighted truth distribution. A fit to the ratio (bottom panel) is used as the prior reweighting factors.

The purity and efficiency are defined as

$$\text{Purity} = 1 - \frac{N_{\text{fake}}}{N_{\text{Reco}}}, \tag{3}$$

$$\text{Efficiency} = 1 - \frac{N_{\text{miss}}}{N_{\text{truth}}}. \tag{4}$$

The purity and efficiency are applied as a bin-by-bin correction to the data spectrum before unfolding and unfolded spectrum after unfolding, respectively. Figure 19 shows the purity, efficiency, and response matrix obtained from the prior-reweighted MC.

Full and half closure tests are performed to validate the overall unfolding procedure. Figure 20 shows the full (left) and half (right) closure tests, with both demonstrating good closure within statistical uncertainties.

The optimal number of iterations is determined by evaluating the iteration stability and the growth of the total uncertainty as a function of the number of iterations. Figure 21 shows the p-value as a function of number of iterations, quantifying the stability between iterations N and $N+1$

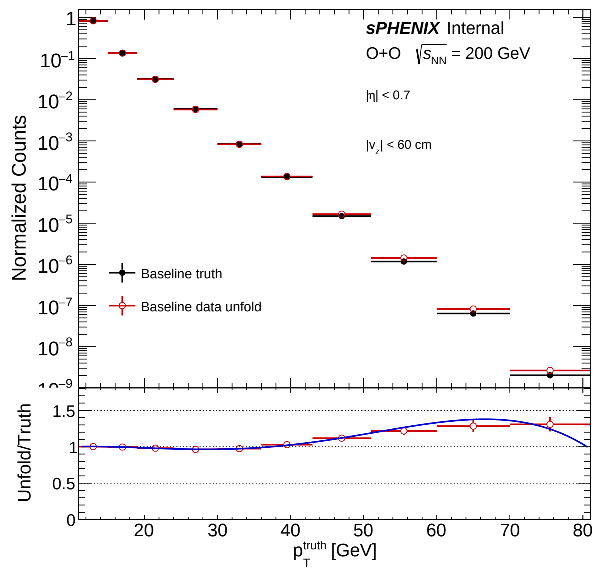

Figure 18: Unfolded data p_T spectrum compared to the baseline unweighted truth spectrum. The bottom panel shows the ratio of the two and used as prior reweighting factors.

for each centrality interval. Figure 22 shows the total uncertainty as a function of the number of iterations, defined as the data statistical uncertainty added in quadrature with the RMS of the response-matrix statistical variation. The latter is obtained from a toy MC procedure, where the response matrix is varied 1000 times within its statistical uncertainty. The optimal number of iterations is chosen to be 2 for the 0–20% and 20–40% centrality intervals, and 1 for the 40–60% and 60–80% intervals.

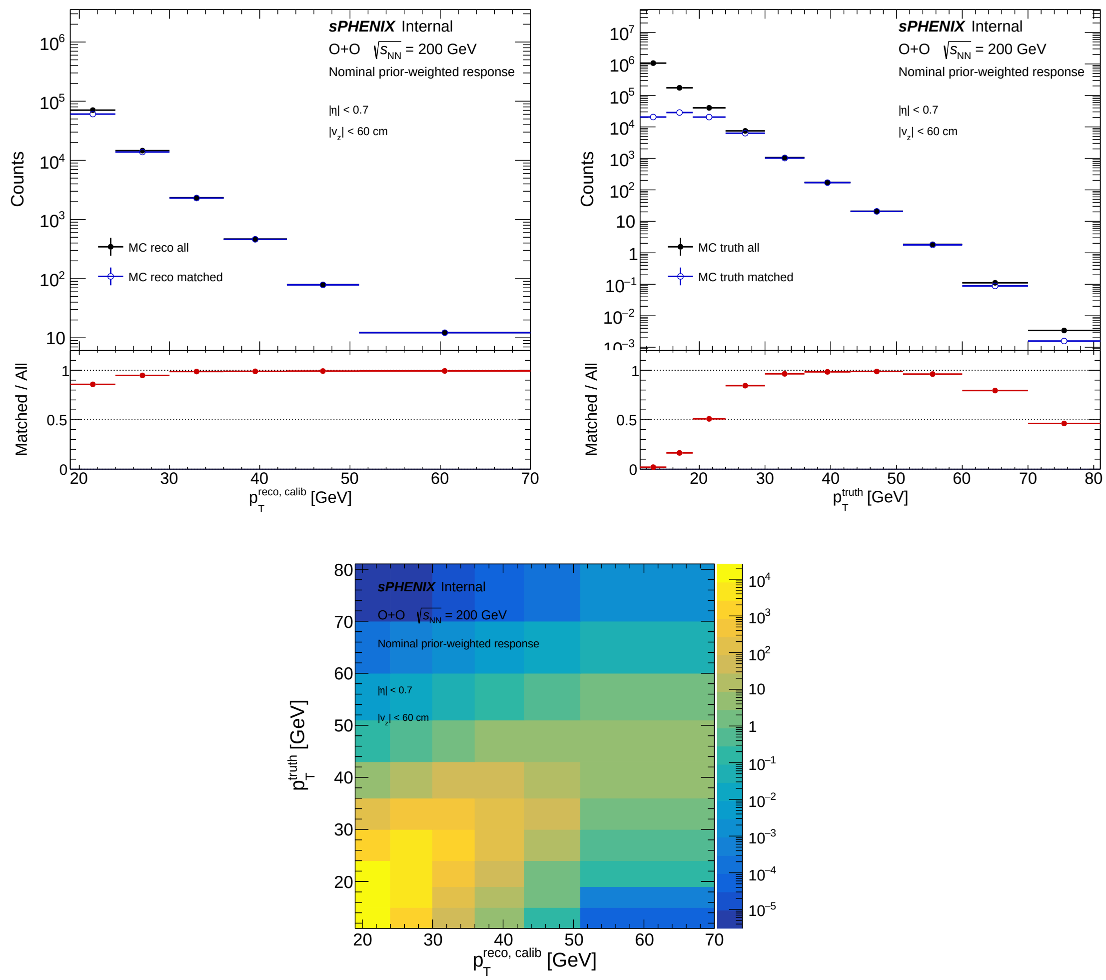

Figure 19: Purity (left), efficiency (right), and response matrix (bottom), evaluated using the prior-reweighted MC sample.

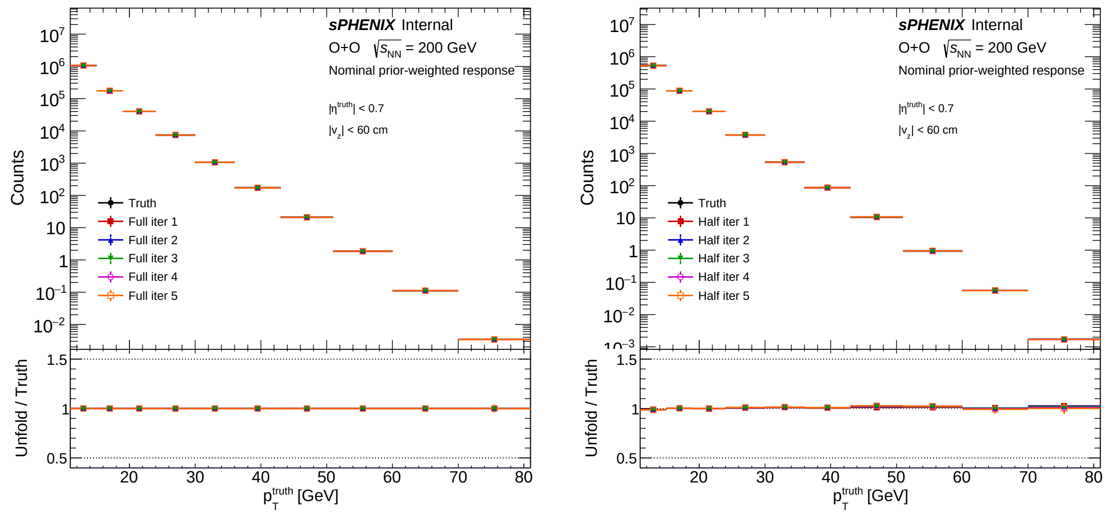

Figure 20: Results of the full (left) and half (right) closure tests.

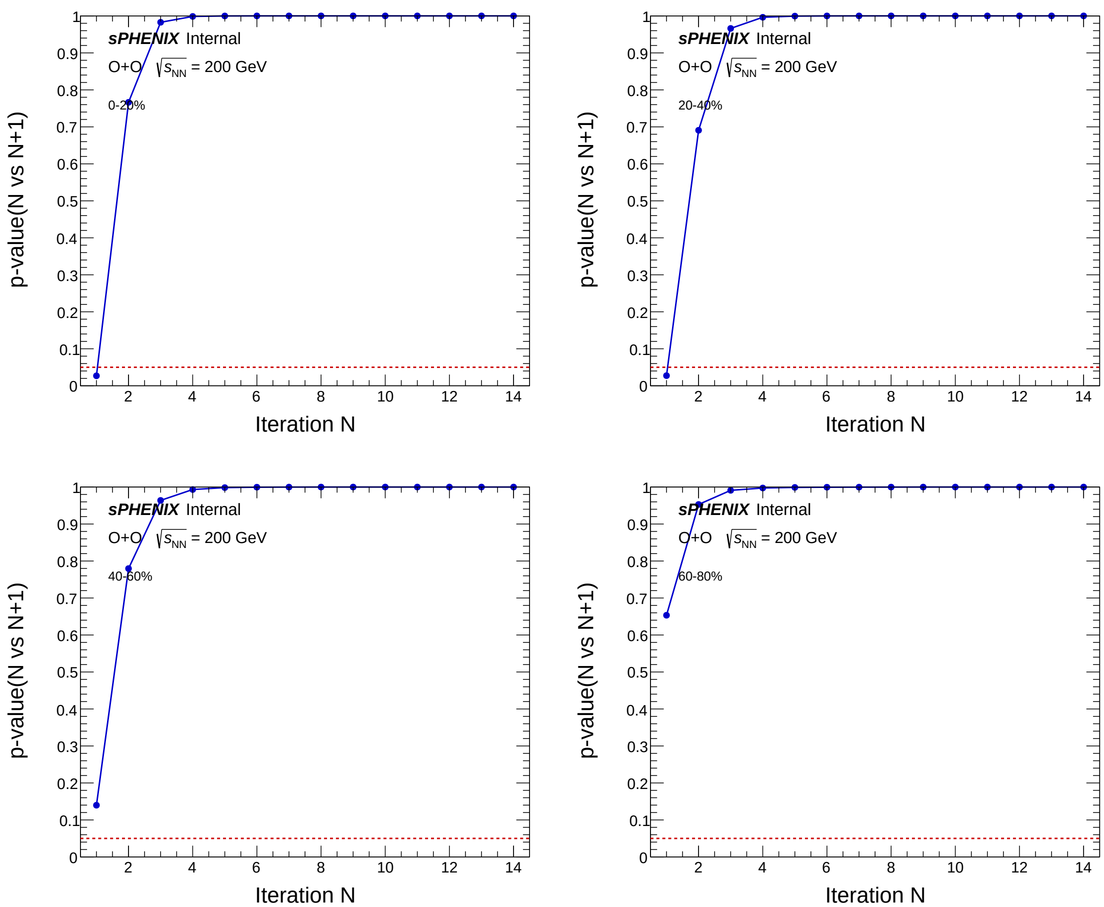

Figure 21: p-value of the iteration-to-iteration stability between iterations N and $N+1$. Panels correspond to the 0–20%, 20–40%, 40–60%, and 60–80% centrality intervals.

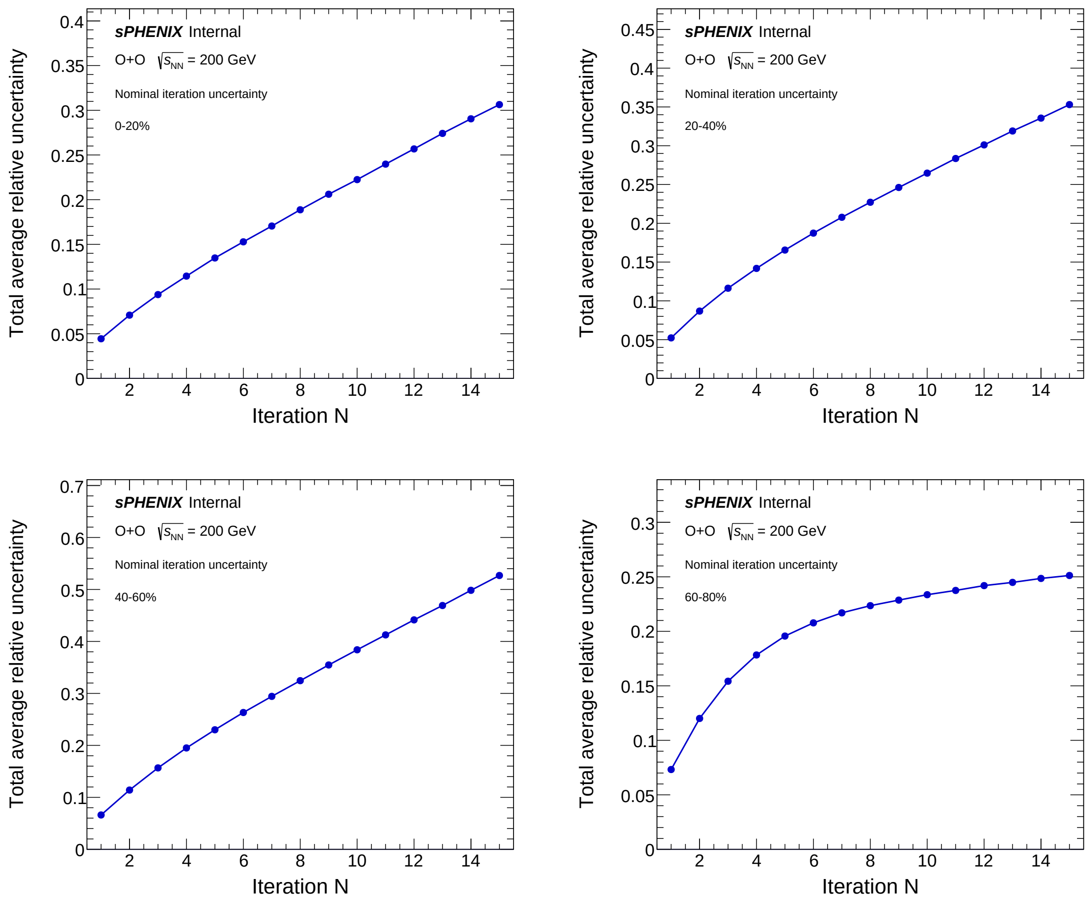

Figure 22: Total uncertainty as a function of the number of iterations. Panels correspond to the 0–20%, 20–40%, 40–60%, and 60–80% centrality intervals.

8 Systematic Uncertainties

Unfolding, MBD modeling, and Global systematics uncertainties not included yet

This section describes the evaluation of the systematic uncertainty for each source considered in this analysis. The sources are:

- JES uncertainty,
- JER variation,
- Unfolding,
- MBD modeling,
- Global (luminosity, Glauber MC variables, centrality calibration).

8.1 JES Uncertainty

To estimate the JES uncertainty, the nominal calibrated p_T of each jet is shifted upward and downward by the uncertainty derived from the *in-situ* JES calibration [11]. The modified spectra are propagated through the full analysis chain, and the difference relative to the nominal result is taken as the JES systematic uncertainty.

8.2 JER Uncertainty

The JER systematic uncertainty arises from the difference in JER between data and MC. The data-driven JER study [10] found that an additional 10% smearing of the MC is required to match the resolution in data. This smearing is varied by $\pm 2\%$, and the difference of the resulting spectra compared to the nominal is assigned as the JER systematic uncertainty.

8.3 Unfolding Uncertainty

Two sources are considered for the unfolding systematic uncertainty. First, the prior reweighting function is varied within its fit uncertainty. The MC is then reweighted with the varied function and propagated through the full analysis chain. Second, the unfolding is repeated with one additional iteration relative to the nominal choice. The differences relative to the nominal result from these two variations are added in quadrature and assigned as the unfolding systematic uncertainty.

8.4 MBD Modeling

To account for possible differences in the modeling of the MBD PMT response between data and MC, the threshold for identifying an MBD PMT hit is varied from 0.5 to 0.4. The difference in the final result compared to the nominal is taken as the MBD modeling systematic uncertainty.

8.5 Global

The uncertainty on N_{coll} values is evaluated by varying the Glauber MC parameters within their uncertainties. The total global uncertainty additionally includes contributions from the MB event selection efficiency.

Some of these uncertainties will cancel out in the R_{CP}

8.6 Total Systematic Uncertainty

The total systematic uncertainty is obtained by adding the individual sources in quadrature, separately for the upward and downward variations. Figure 23 shows a summary of the systematic uncertainties for each measured observable. The different colors represent the individual sources and the black curve represents the total systematic uncertainty.

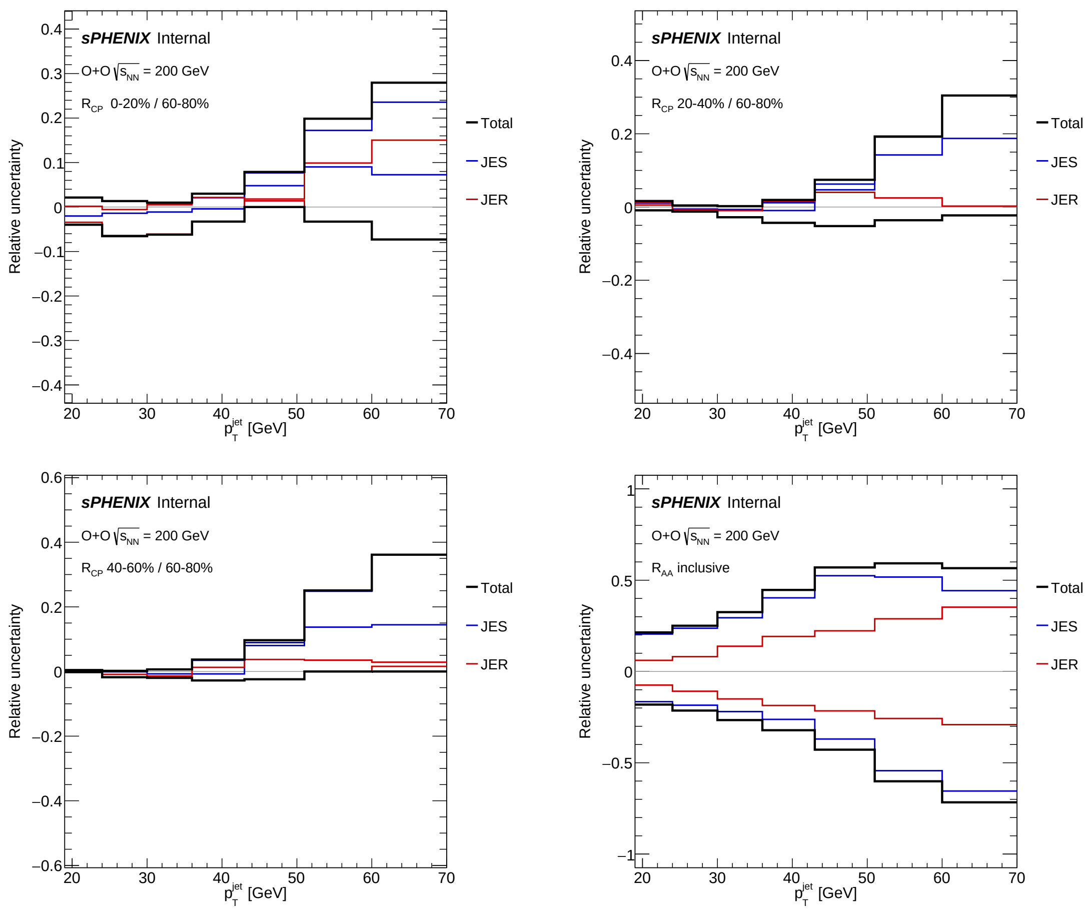

Figure 23: Summary of the systematic uncertainties for each source for R_{CP} and R_{AA}. The black curve represents the total systematic uncertainty.

9 Results

The final R_{CP} and R_{AA} results are presented as a function of jet p_T. Figure 24 shows the R_{CP} and R_{AA} results.

For the R_{AA}, the p+p points are placeholders obtained from PPG09 analyzers. The points will be updated once the analysis gets finalized. For the current plot, all uncertainties (statistical and systematic) are treated as uncorrelated, which is also going to be updated after studying the cancellation of the uncertainties in the ratios.

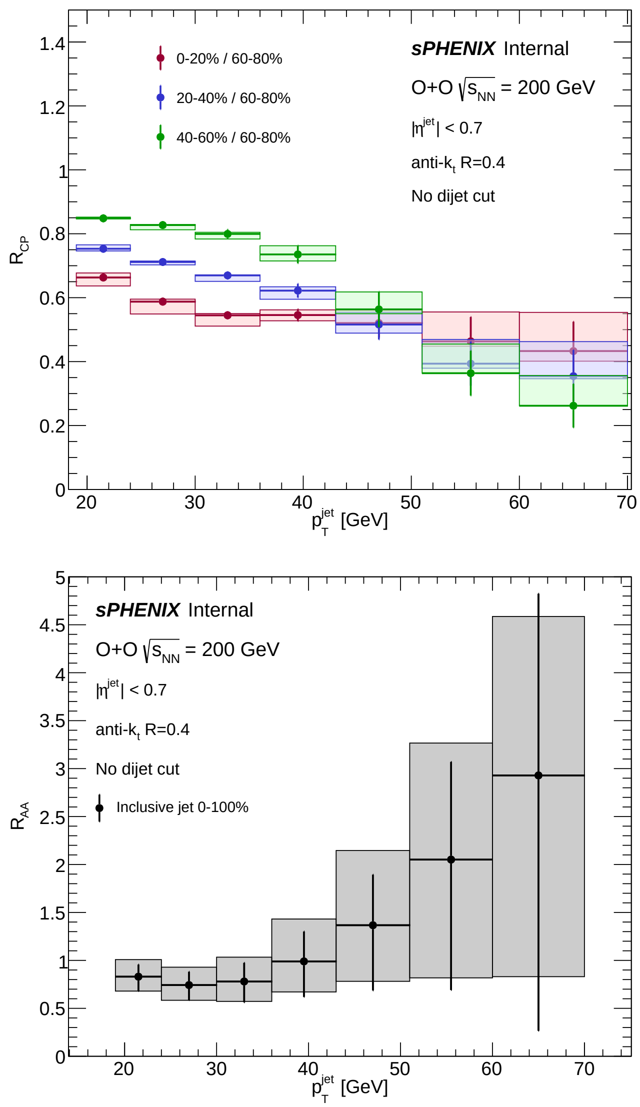

Figure 24: Results for R_{CP} (top) and R_{AA} (bottom) as a function of jet p_T. The colors in the R_{CP} plot represent the different centrality interval selections.

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A Monte Carlo configuration

This appendix summarizes the Pythia 8 configuration used for the p+p Monte Carlo simulations described in Sec. **??**.

Hard-scattering events are generated with Pythia 8.3 [**?**] using the SoftQCD:inelastic = on flag to produce inclusive inelastic events, supplemented by hard QCD $2 \to 2$ processes enabled via HardQCD:all = on with a $\hat{p}_T^{\mathrm{min}}$ threshold set to match the jet p_{T} range of interest. The NNPDF2.3 LO parton distribution function set is used via the LHAPDF6 interface [**?**]. Multiple parton interactions (MPI) and initial- and final-state radiation are enabled with the RHIC-tuned underlying-event parameters from Ref. [**?**]. The configuration cards will be made available on the sPHENIX analysis repository.